Large Language Models in Legal Reasoning and Case Outcome Prediction

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Legal reasoning requires a structured interpretation of facts, statutes, and precedents in novel situations. Recent large language models (LLMs) demonstrate strong general reasoning ability, yet their reliability in the legal field is still unclear. In particular, LLMs can produce legally incorrect arguments, fabricate citations, or exhibit bias. We pursued a project that explores the potential use of advanced LLMs for structured legal reasoning in the prediction of case outcomes and statutory interpretation. This fits option 2 of the assignment: Incorporate ML model in a new Usecase Setting.

CCS Concepts: • **Computing methodologies → Machine learning**; • **Applied computing → Law**.

Additional Key Words and Phrases: Machine Learning, LLM, Legal Reasoning, Prediction, Bias, Law

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1 Introduction

Legal reasoning requires a structured interpretation of facts, statutes, and precedents in novel situations. Recent large language models (LLMs) demonstrate strong general reasoning ability, yet their reliability in the legal field is still unclear. In particular, LLMs can produce legally incorrect arguments, fabricate citations, or exhibit bias. We pursued a project that explores the potential use of advanced LLMs for structured legal reasoning in the prediction of case outcomes and statutory interpretation. This fits option 2 of the assignment: Incorporate ML model in a new Usecase Setting.

Our project required models to (1) extract relevant legal facts, (2) apply doctrinal rules, and (3) produce a structured reasoning trace supporting a predicted outcome. This problem is interesting for three reasons. First, legal reasoning is a high-stakes domain where hallucinations or bias can cause harm. Second, it tests whether recent foundation models truly generalize to structured reasoning tasks beyond conversational fluency. Third, understanding failure modes in legal contexts contributes to research on trustworthy AI and AI governance.

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2 Related Work

This section reflects on prior work relevant to our project across three thematic areas: (1) legal language models and benchmarks, (2) legal judgment prediction, and (3) structured reasoning and reliability in LLMs.

2.1 Legal Language Models and Benchmarks

Early work in legal NLP established that domain-specific pre-training yields meaningful gains over general-purpose models. Chalkidis et al. [6] introduced LEGAL-BERT and pre-training BERT variants. It demonstrated consistent improvements on legal classification tasks such as multi-label topic classification and clause detection. However, LEGAL-BERT is a discriminative encoder and cannot produce explanatory reasoning, making it insufficient for tasks requiring interpretive justification.

Building on this, Chalkidis et al. [7] introduced LexGLUE, a unified benchmark assembling seven legal NLP datasets covering classification tasks such as regulation topic labeling and unfair clause detection. While LexGLUE enables standardized model comparison, all tasks are classification-based and require no open-ended legal reasoning or outcome justification. More recently, Guha et al. [8] presented LegalBench, a benchmark of 162 tasks, finding that even frontier models like GPT-4 struggle most with application and interpretation tasks. Despite its breadth, LegalBench evaluates short-answer or multiple-choice responses and does not assess end-to-end structured reasoning from case facts to predicted outcomes. Hendrycks et al. [9] similarly found in MMLU that few-shot prompted models perform near chance on bar-exam-style law questions, motivating further investigation into prompting strategies for legal tasks.

2.2 Legal Judgment Prediction

Legal judgment prediction (LJP) has been explored as an ML problem since Aletras et al. [10] applied SVMs to predict violation outcomes at the European Court of Human Rights, achieving 79% accuracy. The approach is purely classificatory and provides no reasoning trace. Zhong et al. [12] later showed that even modern reading-comprehension models fail substantially on case-analysis questions requiring multi-hop reasoning across statutes and facts, underscoring that retrieval and classification alone are insufficient for law. Yang et al. [13] addressed this in the Chinese criminal domain. While effective on CAIL2018, this work is limited to statutory civil law systems and does not generalize to common-law precedential reasoning. In the contract analysis domain, Hendrycks et al. [5] introduced CUAD, a dataset of 510 commercial contracts annotated across 41 clause categories, finding that fine-tuned Transformer models perform well on frequent categories but degrade significantly on rare or complex clause types. Koreeda and Manning [4] further framed contract understanding as natural language inference, requiring models to determine whether contractual obligations are entailed or contradicted by a given document, moving closer to relational legal reasoning though still at the sentence level.

2.3 Structured Reasoning and Reliability in LLMs

A similar line of work has focused on eliciting structured reasoning from large language models. Wei et al. [14] demonstrated that chain-of-thought prompting substantially improves LLM performance on arithmetic, commonsense, and symbolic reasoning tasks. Kojima et al. [16] extended this by showing that simply appending "Let's think step by step" to a prompt, without any task-specific examples, elicits multi-step reasoning from sufficiently large models. Wang et al. [15] further improved reliability. These prompting strategies are directly relevant to legal reasoning, where producing intermediate steps mirrors the formal structure of legal analysis. Lewis et al. [17] introduced Retrieval-Augmented Generation (RAG), augmenting LLM inference with retrieved documents to ground outputs in specific precedents or statutes, which is especially relevant given that legal outcomes depend heavily on jurisdiction-specific case law.

Reliability concerns are equally central to legal AI deployment. Lin et al. [18] demonstrated via TruthfulQA that larger models can be more prone to generating confident but false answers, a failure mode with serious consequences in legal contexts such as fabricating case citations or misstating doctrinal standards. Bommasani et al. [19] cataloged broader risks including bias amplification, which is particularly concerning for legal prediction tasks where training data may reflect historical societal disparities. Lippi et al. [20] raised a related concern in their development of CLAUDETTE, a system for detecting unfair clauses in Terms of Service agreements, noting that automated legal analysis can encode and perpetuate existing legal inequities if not carefully evaluated. Taken together, this body of work motivates our focus on not only prediction accuracy but also reasoning faithfulness, hallucination rate, and bias as primary evaluation dimensions.

3 Approach

Each input instance consists of (1) a legal case excerpt describing relevant facts and procedural context, and (2) five candidate legal holdings in multiple-choice format. The model must select the correct holding from among the five candidates. We implement and evaluate two approaches: a majority-class baseline and a fine-tuned Transformer classifier.

3.1 Baseline: Majority Class Predictor

As a lower-bound reference, we implement a majority-class predictor using scikit-learn's `DummyClassifier` with the `most_frequent` strategy. The training set contains 42,509 instances across five answer classes (0–4), with label distribution as follows: class 2 (8,609), class 0 (8,570), class 4 (8,504), class 3 (8,451), and class 1 (8,375). The near-uniform distribution reflects the balanced design of the CaseHOLD dataset. The majority class is label 2, and always predicting this class yields an accuracy of **20.25%** on a held-out 10% stratified test split — consistent with the expected random chance performance of 20% for a five-way classification task.

3.2 Transformer Classifier: ELECTRA-small

Our primary baseline model is a fine-tuned Transformer trained directly on the multiple-choice task. We use `google/electra-small-discriminator` [21] loaded via HuggingFace's `AutoModelForMultipleChoice` interface, which adds a sequence summary and classification head on top of the pre-trained ELECTRA encoder.

Data Preparation. Each instance is formatted as a multiple-choice example by pairing the case excerpt (question) with each of the five candidate holdings individually. The tokenizer encodes each (question, choice) pair separately, producing five input sequences per instance. All sequences are truncated and padded to a maximum length of 300 tokens. The dataset is split 90/10 into training and validation sets using stratified sampling. Due to computational constraints, we sample 1,000 training instances and 300 validation instances for this milestone run, with batches of size 4 for training and 8 for validation.

Training. The model is trained for 5 epochs using the AdamW optimizer [11] with a learning rate of 2×10^{-5}. At each step, the five encoded (question, choice) pairs for a given instance are passed jointly through the model, which produces a logit for each choice. Cross-entropy loss is computed against the ground-truth label and gradients are backpropagated to update all model parameters. Validation accuracy is computed after each epoch by taking the argmax over the five logits and comparing to the true label.

3.3 Planned Extension: LLM with Structured Prompting

In subsequent stages of this project, we hoped to evaluate GPT-4 [3] and Llama 3 [2] under structured zero-shot prompting. However, we were unable to pursue it do to time and monetary restrictions.

3.4 Risks and Mitigation

There are several risks presented:

- Risk 1: Hallucinations. We can mitigate this by constraining to a case database.
- Risk 2: Overfitting. We can mitigate this by ensuring no case excerpt appears in both training and test sets.
- Risk 3: Bias. Analysis remove defendant demographic indicators whenever possible.

3.5 Dataset

We will use the publicly available CaseHOLD dataset from the Harvard Law School Library Innovation Lab. This dataset holds approximately 53,000 multiple-choice legal holding questions. Given a case excerpt and five possible holdings, predict the correct legal holding. The domain is US federal appellate cases and contains cases from 1965-present.

The challenges presented by this dataset are: excerpt might exceed tokens, subtle differences, imbalanced representation, embedded social biases.

3.6 Evaluation Plan

Model performance will be evaluated quantitatively using overall accuracy, macro-averaged F1 score, and per-class accuracy. Our primary baselines will include: (1) a majority-class predictor and (2) a fine-tuned transformer classifier trained directly on the multiple-choice task.

3.7 Computational Resources

Preprocessing, prompting, and evaluation will be implemented in Python using standard libraries. Because we focus primarily on inference rather than large-scale fine-tuning, computational requirements are modest and can be handled on a standard workstation.

4 Data

4.1 Dataset Overview

We use the CaseHOLD dataset [1], introduced by Zheng et al. (2021) and hosted on HuggingFace. CaseHOLD (Case Holdings On Legal Decisions) was constructed from the Harvard Law School Caselaw Access Project corpus, spanning over 3.4 million US federal and state court decisions from 1965 to the present. The dataset frames legal holding identification as a five-way multiple-choice task: given a citing context excerpt from a judicial opinion, the model must identify the correct legal holding of the cited case from among five candidates.

The dataset contains **53,137 instances** in total, split into:

- **Training set:** 42,509 instances
- **Validation set:** 5,314 instances
- **Test set:** 5,314 instances

Each instance contains (1) a `citing_prompt`: a passage from a judicial opinion with a placeholder indicating the location of a parenthetical citation, (2) five candidate holding statements (`holding_0` through `holding_4`), and (3) a ground-truth `label` (0–4) identifying the correct holding. The label distribution in the training set is nearly uniform: class 2 (8,609), class 0 (8,570), class 4 (8,504), class 3 (8,451), and class 1 (8,375), making the majority-class baseline accuracy approximately 20.25%, consistent with random chance on a five-way task.

4.2 Sample Instance

A representative example from the training set is shown below. The citing prompt describes a case involving whether possession of a pipe bomb constitutes a "crime of violence" under federal law:

> *"Felony offenses that involve explosives qualify as 'violent crimes' for purposes of enhancing the sentences of career offenders. See 18 U.S.C. § 924(e)(2)(B)(ii)... Courts have found possession of a bomb to be a crime of violence... See United States v. Newman, 125 F.3d 863 (10th Cir.1997)*

The five candidate holdings are:

(1) *holding that possession of a pipe bomb is a crime of violence for purposes of 18 USC 3142(f)(1)* **[correct]**
(2) *holding that bank robbery by force and violence or intimidation under 18 USC 2113(a) is a crime of violence*
(3) *holding that sexual assault of a child qualified as crime of violence under 18 USC 16*
(4) *holding for the purposes of 18 USC 924(e) that being a felon in possession of a firearm is not a violent felony*
(5) *holding that a court must only look to the statutory definition to determine whether a given offense is by its nature a crime of violence*

4.3 Unique Challenges

CaseHOLD poses several characteristics that make it a particularly challenging benchmark for both fine-tuned classifiers and large language models:

Subtle Doctrinal Distinctions. The incorrect holding candidates are plausible holdings drawn from real cases involving related legal doctrines. As seen in the example above, all five options reference "crime of violence" determinations under different statutory provisions, requiring the model to attend carefully to precise statutory citations and doctrinal nuances rather than surface-level legal topic similarity.

No Reasoning Signal. Unlike datasets that provide chain-of-thought annotations or explanations, CaseHOLD provides only the correct label. This makes it well-suited for evaluating whether models can produce reliable predictions, but means there is no ground truth reasoning trace against which to directly evaluate the structured outputs we elicit from LLMs.

5 Results

5.1 Experimental Setup

We evaluate two models on the CaseHOLD legal holding identification task: a majority-class baseline and a fine-tuned ELECTRA-small Transformer. Due to computational and cost constraints, LLM-based experiments with GPT-4 and Llama 3 were not conducted at this milestone. All experiments use a stratified 90/10 train/validation split of the CaseHOLD training set, with 1,000 training instances and 300 validation instances sampled for the ELECTRA runs.

5.2 Baseline: Majority Class Predictor

The majority-class predictor always predicts label 2, the most frequent class in the training set (8,609 out of 42,509 instances). On the held-out validation split, this yields an accuracy of **20.25%**, consistent with the expected random-chance performance of 20% on a balanced five-way classification task. This result confirms that the dataset is well-balanced across the five holding classes and establishes a meaningful lower bound for model comparison.

5.3 ELECTRA-small Fine-Tuned Classifier

Table 1 reports the training loss and validation accuracy for each epoch of the ELECTRA-small model fine-tuned on 1,000 CaseHOLD training instances over five completed epochs.

5.4 Analysis

Strong Improvement over Baseline. Even with only 1,000 training instances and a relatively small model (ELECTRA-small), the fine-tuned classifier achieves a peak validation accuracy of **69%** at epoch 3 — more than **3.4**× the majority-class baseline of 20.25%. This result strongly confirms that the case excerpt text contains genuine predictive signal for legal holding identification, and that even a small Transformer model can exploit this signal effectively.

Epoch	Training Loss	Validation Accuracy
1	1.6003	66.00%
2	1.3095	66.00%
3	0.9456	69.00%
4	0.6705	64.00%
5	0.4508	62.00%
Majority Class Baseline	–	**20.25%**

Table 1. Training loss and validation accuracy per epoch for ELECTRA-small fine-tuned on CaseHOLD (1,000 training instances, 300 validation instances), compared to the majority-class baseline.

Loss Decreases Steadily; Accuracy is Non-Monotonic. Training loss decreases consistently across all five epochs (1.60 → 1.31 → 0.95 → 0.67), indicating that the model is learning. However, validation accuracy does not follow a monotonic trend: it plateaus at 66% through epochs 1 and 2, peaks at 69% at epoch 3, then drops to 64% at epoch 4. This divergence between falling training loss and declining validation accuracy at epoch 4 is a classic early sign of overfitting — the model begins to memorize the 1,000 training instances rather than generalizing to unseen examples.

Small Training Set as the Primary Bottleneck. The oscillation in validation accuracy and the onset of overfitting by epoch 4 are most likely attributable to the small training sample of 1,000 instances, sub-sampled from the full 42,509-instance training set for computational reasons. With only 300 validation instances, accuracy measurements are also high-variance: a single misclassified example corresponds to a 1% swing. Prior work on CaseHOLD [1] reports that fine-tuned Legal-BERT achieves approximately 75% accuracy on the full training set, suggesting that data scale is the primary factor limiting performance here rather than model capacity.

Comparison to Prior Work. Table 2 situates our results in the context of prior published results on CaseHOLD. Our ELECTRA-small model, trained on only 1,000 instances, already approaches competitive performance relative to larger models trained on the full dataset, which highlights both the difficulty of the task and the effectiveness of Transformer-based representations for legal text even at small scale.

Model	Training Size	Accuracy
Majority Class (ours)	–	20.25%
ELECTRA-small (ours)	1,000	69.00%
BERT-base [1]	Full (42K)	~66%
Legal-BERT [1]	Full (42K)	~75%

Table 2. Comparison of our models against published CaseHOLD baselines. Our ELECTRA-small result is notable given it uses less than 1.2% of the available training data.

5.5 Summary of Findings

Our experiments at this milestone yield three key takeaways. First, the CaseHOLD task is far from trivial: a naive majority-class predictor performs near chance, confirming that legal holding identification requires genuine language understanding. Second, fine-tuning a small Transformer on even a modest subset of training data produces dramatic accuracy gains, validating the utility of pre-trained representations for specialized legal NLP. Third, the early signs of overfitting at epoch 4 and the variance in validation accuracy across epochs indicate that scaling to the full training set will be the most impactful next step. Future work will train on the complete CaseHOLD training split and evaluate structured prompting strategies to compare discriminative and generative approaches to legal holding prediction.

6 Conclusion

This project applied machine learning and language processing to the identification of legal holdings, using the CaseHOLD dataset with more than 53,000 federal case excerpts. We framed the problem as a five-way multiple-choice classification task and evaluated two approaches: a majority-class baseline and a transformer classifier.

Our key findings are as follows. The majority-class baseline achieves 20.25% accuracy, confirming that the near-uniform label distribution of CaseHOLD makes naive prediction strategies ineffective and that the task demands genuine language understanding. Fine-tuning ELECTRA-small on just 1,000 training instances — less than 1.2% of the available training data — yields a peak validation accuracy of 69%, representing a 3.4× improvement over the baseline. This result demonstrates that pre-trained Transformer representations capture legally meaningful features even without domain-specific pretraining or large amounts of labeled data. At the same time, the non-monotonic validation accuracy trajectory and the divergence between falling training loss and declining validation accuracy at epoch 4 suggest that the small training sample size is the primary bottleneck, and that the model begins to overfit before fully generalizing to the legal holding identification task.

Several lessons emerged from this work. Legal NLP is a genuinely difficult domain: the subtle distinctions between candidate holdings — which differ not in legal topic but in precise statutory provision, jurisdictional scope, and doctrinal framing — demand fine-grained contextual reasoning that simple classification models only partially capture. The CaseHOLD task is also notable for what it does not provide: there are no reasoning annotations, no explanation of why a holding is correct, and no intermediate inferential steps. This absence makes it difficult to diagnose model failures or assess whether a correct prediction reflects genuine legal reasoning or superficial pattern matching.

Several directions remain open for future work. Most immediately, training on the full 42,509-instance training set rather than the 1,000-instance subsample used here is expected to yield substantial accuracy gains and reduce the overfitting observed at epoch 4. Replacing ELECTRA-small with a larger or domain-adapted model such as Legal-BERT [6] or DeBERTa [5] would further close the gap with state-of-the-art published results on CaseHOLD. Beyond discriminative classifiers, evaluating structured zero-shot prompting of large language models such as GPT-4 [3] and Llama 3 [2] remains a compelling direction: rather than simply predicting a label, such models could be prompted to produce an explicit IRAC-structured reasoning trace — extracting relevant facts, identifying the applicable legal rule, applying it to the facts, and arriving at a predicted holding — enabling a richer evaluation of both prediction accuracy and reasoning faithfulness. Measuring hallucination rates, citation accuracy, and doctrinal coherence in generated reasoning traces would contribute to the broader question of whether current LLMs are reliable enough for deployment in high-stakes legal contexts. Finally, analyzing model performance disaggregated by legal domain, case era, and circuit jurisdiction could reveal systematic failure modes and biases that aggregate accuracy metrics obscure.

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